

BOONE, IA, USA

WMO: 725486

Lat: 42.049N

Lon: 93.848W

Elev: 354

StdP: 97.15

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04906

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-17.8	-25.1	0.4	-20.8	-22.4	0.5	-17.6	13.3	-9.1	11.9	-6.9	4.3	340	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	32.8	24.6	31.6	24.1	30.0	23.1	26.7	30.8	25.5	29.4	24.4	28.4	4.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	21.3	29.4	24.1	19.9	28.0	22.9	18.4	27.0	85.8	30.7	80.9	29.7	75.5	28.6	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.5	9.1	-25.3	35.1	4.0	1.6	-28.2	36.3	-30.5	37.2	-32.7	38.1	-35.6	39.3		
(5)			-25.1	28.3	3.7	1.1	-27.7	29.1	-29.9	29.8	-32.0	30.4	-34.6	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-5.7	-3.5	2.7	9.8	16.2	21.5	23.4	22.4	18.3	11.2	3.7	-2.9	(6)
(7)		DBStd	11.58	6.94	6.89	6.98	5.37	4.89	3.61	3.09	2.91	4.42	5.39	6.15	6.21	(7)
(8)		HDD10.0	1834	486	378	242	67	7	0	0	0	2	49	204	399	(8)
(9)		HDD18.3	3597	744	611	484	259	101	13	2	3	54	229	440	657	(9)
(10)		CDD10.0	1773	0	1	17	62	198	344	415	383	250	87	14	1	(10)
(11)		CDD18.3	494	0	0	1	4	34	107	159	128	53	8	0	0	(11)
(12)		CDH23.3	4384	0	0	7	66	333	950	1442	1018	476	88	5	0	(12)
(13)		CDH26.7	1452	0	0	0	14	106	326	514	322	146	24	1	0	(13)
(14)	Wind	WSAvg	4.4	4.8	4.9	5.2	5.5	4.9	3.9	3.1	2.9	3.5	4.4	4.7	4.6	(14)
(15)	Precipitation	PrecAvg	913	25	28	59	94	125	130	126	121	75	54	48	29	(15)
(16)		PrecMax	1400	52	49	131	204	215	289	372	284	174	145	112	57	(16)
(17)		PrecMin	631	9	3	3	21	34	21	41	24	25	8	0	0	(17)
(18)		PrecStd	175	11	14	36	50	50	69	78	67	38	34	32	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	16.3	24.1	28.7	32.5	33.7	35.0	34.1	32.7	30.0	23.4	15.1	(19)
(20)			MCWB	6.3	11.3	15.8	18.1	21.3	24.3	26.8	24.8	22.6	21.1	14.8	11.9	(20)
(21)		2%	DB	7.9	12.1	20.0	25.0	29.2	32.2	32.8	32.1	30.7	25.8	18.9	10.8	(21)
(22)			MCWB	4.4	8.8	13.8	15.7	19.6	23.9	25.7	25.1	22.0	17.8	12.9	8.0	(22)
(23)		5%	DB	5.2	8.0	16.3	22.2	27.2	30.2	31.3	30.0	27.9	22.7	15.8	7.6	(23)
(24)			MCWB	2.8	5.6	12.4	14.0	18.4	22.4	25.1	24.3	20.6	16.2	10.9	5.1	(24)
(25)		10%	DB	2.7	5.2	12.8	19.0	24.6	28.4	29.8	27.9	26.3	20.1	12.7	5.0	(25)
(26)			MCWB	1.2	3.1	9.1	12.5	17.3	21.7	24.2	22.9	19.5	14.7	8.7	2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	12.1	17.3	19.4	23.5	26.9	28.8	28.0	25.0	21.9	15.7	12.8	(27)
(28)			MCD B	11.5	15.4	22.3	25.7	29.0	31.2	33.2	32.2	29.1	26.5	20.8	14.7	(28)
(29)		2%	WB	4.9	8.6	15.0	17.0	21.4	25.5	27.1	26.2	23.7	19.6	13.9	8.5	(29)
(30)			MCD B	7.9	11.1	18.2	22.6	27.4	29.9	31.2	30.1	28.1	23.9	17.9	9.5	(30)
(31)		5%	WB	3.0	5.8	12.6	15.2	19.9	24.0	25.9	25.0	22.6	17.5	11.4	5.2	(31)
(32)			MCD B	5.1	8.3	16.3	20.3	24.6	28.4	29.6	28.8	26.5	20.8	15.1	7.6	(32)
(33)		10%	WB	1.3	3.4	9.5	13.3	18.7	22.8	24.8	23.7	21.1	15.3	9.0	2.9	(33)
(34)			MCD B	2.7	5.2	12.9	18.1	23.0	27.0	28.5	27.3	24.5	19.2	12.5	4.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	9.8	11.4	10.9	10.5	10.2	10.0	11.9	11.1	9.6	8.0	(35)
(36)			MCD BR	11.2	11.7	14.6	17.1	14.1	12.3	11.1	11.6	13.9	14.8	14.3	11.0	(36)
(37)			MCWBR	8.4	8.6	9.2	9.2	6.6	6.1	6.1	6.5	7.0	8.1	8.9	8.6	(37)
(38)		5% WB	MCD BR	10.5	11.5	13.7	13.9	11.8	10.8	10.2	10.3	11.4	11.9	13.0	10.2	(38)
(39)			MCWBR	8.1	8.8	9.1	8.8	6.7	6.6	6.7	6.7	6.6	8.1	9.0	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.302	0.323	0.379	0.408	0.421	0.435	0.419	0.388	0.341	0.297	0.284	(40)	
(41)		taud	2.421	2.378	2.421	2.284	2.273	2.282	2.223	2.275	2.318	2.443	2.507	2.446	(41)	
(42)		Ebn at Noon	867	904	926	889	869	855	838	841	842	848	849	837	(42)	
(43)		Edh at Noon	77	95	103	127	132	131	138	127	113	89	71	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.90	2.92	3.79	4.75	5.61	6.23	6.35	5.46	4.52	3.04	2.06	1.56	(44)	
(45)		RadStd	0.19	0.21	0.37	0.39	0.43	0.55	0.42	0.36	0.32	0.37	0.16	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page